

Review 1

General comments: The manuscript proposes a multi-segment fusion architecture for pediatric bone age estimation using four anatomical regions—pinky, middle finger, thumb, and wrist—and compares F-ResNet50, F-VGG16, F-DenseNet121, and F-InceptionV3, with external validation on 100 Mexican clinical radiographs.

The topic is clinically relevant, especially because the paper emphasizes demographic shift and the need for local validation in Mexican pediatric populations.

The manuscript has a potentially useful contribution, but the current version has major methodological, reporting, and interpretation weaknesses that prevent acceptance without substantial revision.

Response: The authors would like to thank the reviewer for the thorough and constructive evaluation of our manuscript. We greatly appreciate the time devoted to this review, as well as the detailed comments and suggestions, which have substantially improved the manuscript's clarity, transparency, and scientific rigor. Below, we address each concern point by point.

- **Strengths**

Addresses an important limitation in bone age estimation: poor external generalization across demographic and clinical populations.

Includes an external Mexican validation set, which is a meaningful addition beyond standard RSNA-only evaluation.

The segment-based design is clinically motivated and attempts to preserve anatomical information rather than relying only on whole-hand radiographs.

The comparison across multiple backbones under a shared fusion framework is useful for isolating the effect of architecture choice.

The paper clearly reports that internal validation performance can substantially underestimate true test and external-validation error.

Response: The authors thank the reviewer for this positive assessment of the manuscript's strengths, which we have worked to preserve and reinforce throughout the revision.

- **Major concerns**

Decision-driving issue: the claimed contribution is weakened by test-set performance.

F-DenseNet121 achieves 5.77 months MAE on validation but 13.70 months on the independent RSNA 20% test set, showing a large generalization gap.

The comparison to prior methods is unfavorable: reported state-of-the-art RSNA MAEs are approximately 4.2–6.2 months, whereas the proposed model reports 13.70 months on the RSNA test set.

The manuscript should not describe the method as “competitive” without carefully qualifying that the external-validation design is valuable, but the RSNA test performance is substantially worse than leading published systems.

Response: According to the reviewer's suggestion, this point has been clarified in the new version of the manuscript. We have revised the discussion of Table 10 (Section 3, “Results and Discussion”) to explicitly state that the model's performance on the independent RSNA hold-out test set (13.70 months) remains substantially higher than the state-of-the-art figures reported by Halabi et al. (2019) and Ren et al. (2019), and that this

gap should be interpreted as a current limitation of the proposed method rather than evidence of overall competitive performance. The word “competitive” is now used only in reference to internal validation performance, and this qualification has also been added to the new “3.5 Limitations” section.

The validation protocol is unclear and potentially problematic.

The manuscript states that RSNA is split into training, validation, and test sets, with a 20% test split, but it does not provide exact sample counts by split, patient-level separation details, random seed, stratification by age/sex, or whether any tuning decisions used validation/test feedback.

Response: The authors believe this point is very important; in the revised manuscript (Section 2.1, Database) now explicitly states that the RSNA dataset was partitioned into an independent 20% hold-out test set (2,847 radiographs), reserved prior to model development, and the remaining 11,389 radiographs used for training and validation, with partitioning performed to avoid patient-level overlap across subsets.

The difference between “validation,” “independent 20% RSNA hold-out test,” and any official RSNA challenge split must be clarified.

Response: The authors have attended the reviewer’s comment. As clarified in the revised Section 2.1, from the total RSNA dataset, 20% of the images were held out independently before any development to form the “independent 20% RSNA hold-out test” set, used only for the final evaluation reported in Table 8. The remaining 80% of the RSNA images were internal validation split, obtained by separating 20% of the training data, used for model adjustment and hyperparameter tuning during training, and is therefore distinct from the independent hold-out test set. This study uses the official RSNA challenge test-set labels (which were released separately by the organizers); this distribution has been made explicit in the revised text to avoid confusion with the official RSNA challenge split.

The paper should explicitly state whether segmentation models, preprocessing thresholds, augmentation settings, and fusion hyperparameters were selected before test evaluation.

Response: The authors have attended reviewer’s comment. A new sentence has been added at the end of Section 2.5 (Experimentation) explicitly stating that all hyperparameters (Tables 2-3) and preprocessing thresholds (Section 2.2) were fixed prior to any evaluation on the independent 20% RSNA test set or the Mexican external dataset, both which were held put and not used for model selection or tuning decisions.

External Mexican dataset is too weakly described.

The paper reports only 100 Mexican radiographs and basic demographic plots.

Response: The authors acknowledge this limitation. As now stated in the revised Section 2.1 and in the new “3.5 Limitations” section, the external Mexican dataset, while a valuable contribution, is limited to 100 radiographs; a larger, prospectively collected and more thoroughly characterized clinical cohort is identified as necessary future work to draw firmer conclusions about generalization to the Mexican pediatric population.

Missing details include age range by sex, acquisition equipment, clinical source, inclusion/exclusion criteria, left/right hand consistency, disease indications, annotation process, rater expertise, inter-rater variability, and ethics/IRB approval.

Response: The authors have attended the reviewer's comment. Section 2.1 (Database) has been expanded to describe the age range, sex distribution, acquisition site, imaging equipment, and anonymization procedure for the Mexican dataset.

Ground truth for the Mexican dataset is not adequately described: it is unclear whether "chronological age," clinician-assigned bone age, TW3 score, GP atlas reading, or another reference standard was used.

Response: The authors have attended the reviewer's comment. The revised Section 2.1 now states that ground-truth bone age for the Mexican dataset was assigned using the TW3 reference method by qualified raters, with the specific rater credentials to be confirmed and included before resubmission.

The segmentation stage needs stronger evidence.

The paper reports Dice, IoU, and accuracy for segmentation and selects MobileNetV2 for efficiency despite DenseNet121 having a higher Dice score.

Response:

Los experimentos con otras arquitecturas únicamente se diseñaron para validar que la arquitectura seleccionada (MobileNetV2) se colocaba entre las mejores, no para asegurar el mejor.

Los experimentos se crearon para validar que la segmentación fuese correcta, aunque no se buscó la más óptima sino obtener los segmentos a como se necesitan para el procesamiento y entrenamiento de los modelos.

It is unclear how segmentation ground-truth masks were produced, how many images were manually annotated, whether masks were reviewed by clinical experts, and whether segmentation validation used RSNA only or also Mexican images.

Response:

Producidas con software labelMe, 200 imágenes etiquetadas manualmente encerrando las 13 regiones del método TW3, si fue revisada por el experto de grupo de trabajo del artículo (Daniel). La validación únicamente se hizo con RSNA dataset.

Pixel accuracy is not very informative for imbalanced segmentation masks; Dice/IoU should be primary, with confidence intervals.

Response:

Quizá agregar más detalle a la sección 3.1? Porque si está Dice /IoU reportado

The fusion method requires ablation and statistical testing.

The paper asserts that fusion improves performance, but the evidence is not sufficient unless directly compared with whole-hand single-input models trained under the same split, preprocessing, optimizer, and augmentation conditions.

Response:

Se realizaron otros experimentos en donde se usó la arquitectura fusionada en comparación con simple CNN en donde se demuestra la diferencia

3 3	●	simple_cnn	N/A	112×12	spatial	raw	24 -2 16	✓	✓	39 .2 m	35 .9 m
3 4	●	backbone	DenseNet 121	112×12	spatial	raw	24 -2 16	✓	✓	14 .6 m	17 .6 m

The additional fusion comparison in Section 3.3 is promising, but the dataset settings “trimmed, balanced, and full” are not defined clearly enough.

Response:

Debido a la distribución del dataset RSNA se optó por experimentar de 3 formas.

- (Trimmed) Una recortando las colas de la distribución que correspondía a edades muy poco representadas.
- (Balanced) Usando un threshold de al menos 50 imágenes por edad para representatividad y descartando aquellas edades con menos de 50 imágenes
- (Full) El dataset completo sin ningún filtro o recorte

Statistical uncertainty is missing. MAE values should include confidence intervals or bootstrapped intervals, especially for the Mexican dataset with only 100 cases.

Response:

Hacer bootstrapped intervals

Some interpretations are overextended.

The manuscript states that InceptionV3 robustness may come from multi-scale features identifying stable morphological patterns.

Response:

Lo quitamos? Pendiente

This is plausible but not demonstrated. The claim should be framed as a hypothesis unless supported by feature-level analysis, saliency consistency, subgroup tests, or error decomposition.

Response:

Este se refiere al mismo de arriba.

Saliency maps are presented as anatomical coherence evidence, but one example is not enough to support interpretability claims.

Response:

Agregar más ejemplos?
Explicar cómo funcionan los mapas de saliencia
Agregar 3 imágenes aleatorias

Reporting and reproducibility are insufficient.

Optimizer, loss function definition, learning-rate schedule, early stopping, initialization, hardware, software framework, preprocessing implementation, and code/data availability are not fully specified.

Response:

Todo esto sí lo puedo agregar al paper.

Image size is reported as 112×112 px for prediction, but segmentation input is described as 224×244×3; this discrepancy needs explanation.

Response: The authors have attended the reviewer's comment. The typo in Section 2.3 has been corrected (224x224x3), and a clarifying sentence has been added explaining that the segmentation-stage resolution intentionally differs from the 112x112x3 px input used by the boundary detail for mask generation while the reduced resolution lowers computational cost for the four parallel prediction branches.

Table 7 appears to contain a formatting error in the F-InceptionV3 parameter count: "89,312261" should be checked.

Response: This mistake has been corrected.

- **Minor concerns**

The abstract contains grammar and clarity problems, including "where promotes," "suggest difference among public and real clinical datasets," and inconsistent MAE reporting of 5.76 vs. 5.77 months.

Response: The authors have attended the reviewer's comment. The abstract has been rewritten to correct the grammar and clarify problems identified, including the "where promotes" construction and the "suggest difference" phrasing.

Several sections use repetitive wording such as "under such circumstances," which should be reduced.

Response: The authors have attended the reviewer's comment. The repeated phrase "under such circumstances" has been replaced with transition phrases (e.g., "accordingly", "as a result", "consequently", "building on this rationale", "following this rationale", and "given this design") throughout the Introduction and Materials and Methods sections.

The manuscript repeatedly uses "chronological bone age," which is conceptually confusing; chronological age and bone age should be distinguished.

Response: The reviewer is right; the confusing term “chronological bone age” has been corrected to “chronological age” in Section 3.2 (Figure 7 description), which correctly distinguishes it from the predicted bone age.

The “5. Patents” section appears to contain author contributions rather than patents.

Response: The authors have attended to the reviewer's comment. The "5. Patents" section, which erroneously contained the author contributions, has been removed, and the Author Contributions statement now appears separately, as intended by the journal template.

The abbreviation list incorrectly defines “AE” as “Multidisciplinary Digital Publishing Institute”; AE appears to mean Absolute Error.

Response: The authors have attended the reviewer’s comment. The abbreviation table has been corrected so that “AE” is now defined as “Absolute Error” instead of “Multidisciplinary Digital Publishing Institute”.

The title emphasizes F-DenseNet121, but Mexican validation shows F-InceptionV3 has the best external MAE. The title, abstract, and conclusions should better reflect this result.

Response:

- **Required revisions before reconsideration**

Clarify all dataset splits, sample counts, stratification, and leakage-prevention procedures.

Fully describe Mexican dataset collection, ethics approval, annotation/reference standard, and clinical characteristics.

Add confidence intervals and statistical comparisons for MAE, especially on the Mexican set.

Include a same-protocol whole-hand CNN baseline and a no-fusion baseline.

Define the fusion ablation settings and report them in a complete table.

Temper claims about competitiveness, interpretability, and robustness.

Correct formatting, terminology, abbreviation, and section-label errors.

Improve English throughout for precision and readability.

- **Final decision**

Major Revision.

The manuscript presents an interesting clinical and scientific question with an important validation aspect; however, at present, the existing evidence cannot support the bold claims

made. Reconsideration for acceptance can take place only after further elucidation of the experimental design, improved analysis of the ablation, and statistical uncertainties.

Review 2

General comments: This paper presents a multi-region fusion architecture that uses the TW3 protocol to estimate bone age based on four distinct anatomical locations (the thumb, middle finger, pinky finger, and wrist). It also includes a comparative analysis and external validation using a clinical data set from Mexico. There are a number of issues that need to be resolved even if the research addresses a pertinent clinical issue and offers some helpful information regarding external validation.

General response: The authors would like to thank you for the careful and constructive review of our manuscript. We greatly appreciate the time devoted to this task, as well as the recommendations and suggested changes, which have significantly improved the manuscript. Below we address each comment point by point, with corresponding changes marked in red font in the revised manuscript.

Comments 1. The authors should reframe their contribution to emphasize the systematic evaluation framework and external validation methodology rather than claiming architectural novelty. The value lies in the rigorous comparative analysis and demographic generalization study.

Response 1:

Comments 2. Why choose exactly 4 regions? Why not 5 or 6? Why 112×112 image size when most bone age models use larger inputs (e.g., 224×224)? Why 15/20/10 epochs for segment/fusion/fine-tuning stages? This seems arbitrary without justification or ablation studies.

Response 2:

Las 4 regiones se seleccionaron para abarcar las 13 secciones del método de TW3, la ubicación de los grupos de regiones permite reducir la cantidad de modelos de entrenamiento además de su fácil localización.

La resolución de la imagen se decidió para reducir el tiempo de procesamiento entre experimentos y dar cabida a más combinaciones de configuración.

La cantidad de épocas se eligió al verificar que ningún experimento lograba una mejora significativa después de esa cantidad de épocas.

Comments 3. No statistical tests (e.g., paired t-tests, ANOVA) are reported to determine if performance differences between architectures are significant. The difference between F-InceptionV3 (15.98 months) and F-DenseNet121 (16.30 months) on the Mexican dataset may not be statistically significant.

Response 3:

Hacer Prueba pareada de todos los F-Model

Comments 4. The 20% hold-out test set results (Table 8) show **dramatically** different MAE values than validation results. The paper mentions this as a "generalization gap" but does not explain why this gap exists (e.g., distribution differences in the test split?)

Response 4:

Al no ser una base de datos distribuída de forma uniforme, el 20% hold-out test contiene diferencias en grupos de edad comparado con el dataset de entrenamiento, lo que permite que el modelo tenga problemas para generalizar sobre edades que nunca vio durante entrenamiento.

Comments 5. Why DenseNet121 over other architectures? The paper cites feature reuse and gradient flow, but this is generic - how do these properties specifically benefit bone age estimation?

Response 5:

Podemos justificar con la reducción de MAE y la cantidad e parámetros entrenables?